

# Financial Literacy Chatbot Using RAG and LLMs

FinBuddy

## Problem Selection & Justification

- Financial illiteracy affects millions globally, leading to poor decisions around budgeting, investing, and credit. While information is available online, it is often complex and fragmented.
- This project aims to build an intelligent chatbot that improves financial literacy by answering user queries in simple terms using Retrieval-Augmented Generation (RAG) and a fine-tuned Large Language Model (LLM). It will help users understand financial concepts and make informed decisions.

## Dataset

- **Sources:** Glossary terms and definitions from [Investopedia](https://www.investopedia.com), CFPB, and SEC Investor.gov.

## Network Architecture

- We will use a **RAG pipeline**, combining:

**Retriever:** Dense vector search using sentence-transformer embeddings and FAISS.

**Generator:** A fine-tuned model to generate domain-specific, grounded answers.

- The LLM will be trained to explain financial terms in simple, accurate language.

## Framework

- Implemented using Hugging Face Transformers, LangChain, and Streamlit.

## Performance Evaluation

- **Primary Metric:** Human-evaluated clarity and correctness of generated answers.
- **Secondary Metrics:** Precision@k (retriever), BLEU/ROUGE (generator), response latency, and user feedback.